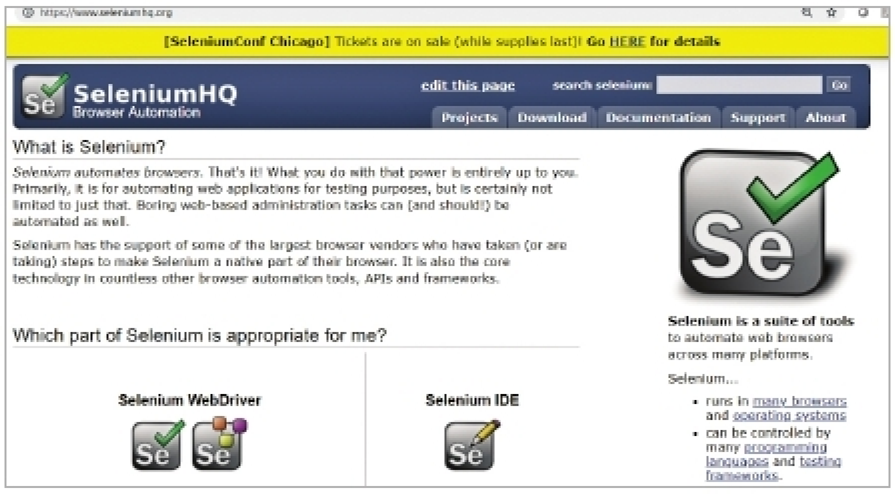

Figure 1: Selenium

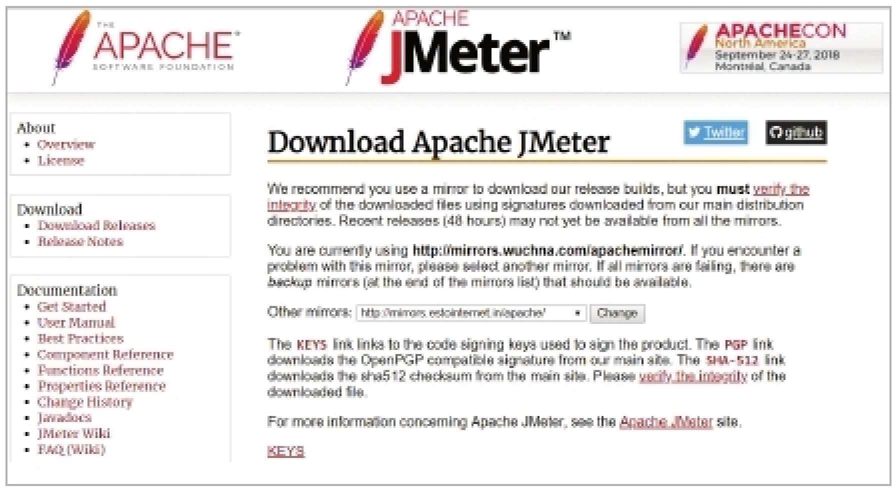

Figure 3: Apache JMeter

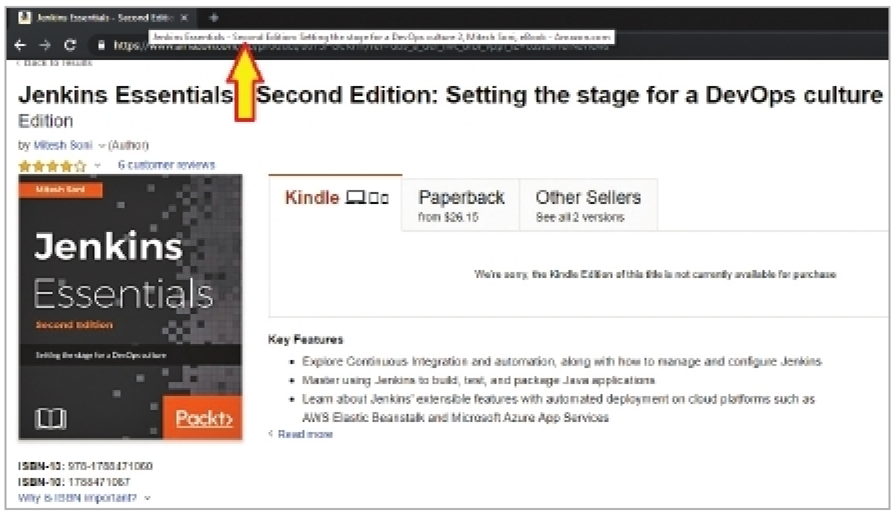

Figure 2: Functional testing with Selenium

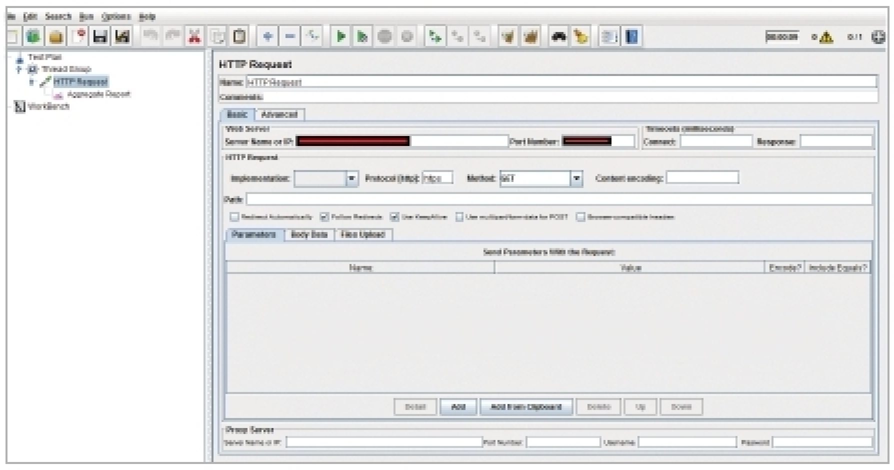

Figure 4: Apache JMeter dashboard

Name	Selenium
Objective	Functional testing for Web applications
Licence	Apache Licence 2.0
Stable release	3.14.0
Website	*https://www.seleniumhq.org/*
Supported operating systems	Cross-platform
Written in	Java
Support for integra-tion with continuous integration tools	Selenium test cases can be written in Java and can be easily integrated with CI tools such as Jenkins as a Maven based project. Selenium tests can also be written in Scala, C#, Groovy, Perl, PHP, Python and Ruby.

So let's look at an example of how Selenium is used for sample functional tests.

- The title of the page must contain specific text or a sub-heading available in the book page available in Amazon (as shown in Figure 2).
- We need to identify the input for the function test we are going to execute.
  - We need to check whether the specific Web page *https://www.amazon.com/gp/product/B073PGCKHY* has a specific header set or not.
  - Here we know the input for our functional test—URL of a Web page and header string; so we have our test input data available.
- We need to get the expected results with the test we have selected. Execute the functional test case and save the outcome.
  - Eclipse is the IDE.
  - Go to the Eclipse marketplace. Install the Maven Integration for Eclipse plugin.
  - Create a Maven project.
  - Install Maven properly. If Maven is behind a proxy server, configure the proxy details in *conf.xml*, available in the *M2_HOME* directory.
  - Add Maven, Selenium, TestNG and JUnit dependencies to *POM.XML*.
  - Install the TestNG plugin and write the TestNG class using the Eclipse IDE.
  - Right-click on the *test* file, and click on *TestNG* to convert to *TestNG*. This will create a *testing.xml* file that has details about the test suite. Next, right-click on *Project* and click on *Run Configurations*. Then right-click on *TestNG* and click on *New*. Provide the project name and select *testing.xml* in the suite.
  - Create the *TestNG* class under the *test* folder. Select the location, suite name, and class name:

```
<?xml version="1.0" encoding="UTF-8"?>
    <!DOCTYPE suite SYSTEM "http://testng.org/testng-
```